

Summary - Heterogeneous Multi Agent

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1. Overall Summary

This paper addresses the challenge of dynamically optimizing cache storage and content fetching in Cognitive Radio Networks (CRNs), a complex task due to unpredictable content popularity and primary user spectrum occupation. Existing Deep Reinforcement Learning (DRL) methods often fall short by treating multi-cache systems as homogeneous agents or a single global agent, neglecting the autonomous and often conflicting objectives of users (minimizing delay) and cache servers (maximizing revenue). To overcome this, the authors propose a novel heterogeneous Multi-Agent Deep Deterministic Policy Gradient (MADDPG) framework. In this model, users and Secondary Base Stations (SBSs) are treated as two distinct types of agents. SBSs learn to optimize the cached fraction of each content to maximize storage efficiency, while users learn to select the best SBS for content fetching to minimize delivery delay. The MADDPG architecture employs centralized training for stability and decentralized execution for reduced communication overhead. Numerical simulations demonstrate that this heterogeneous MADDPG significantly outperforms single-agent or homogeneous DRL approaches, including DDPG, PPO, TD3, and homogeneous MADDPG variants, achieving up to a 40% improvement in total system reward by effectively balancing the cooperative and competitive interactions between agents.

2. Main Idea

The central concept is to propose a heterogeneous multi-agent deep deterministic policy gradient (MADDPG) approach for cooperative caching in Cognitive Radio Networks (CRNs). This approach models both users and Secondary Base Stations (SBSs) as distinct types of autonomous agents that learn to cooperate and compete to simultaneously reduce user content delivery delay and enhance the cache efficiency of SBSs.

3. Problem Being Solved

This paper aims to solve the problem of dynamically optimizing cache storage and content fetching strategies in Cognitive Radio Networks (CRNs). This is a non-trivial task due to the unpredictable nature of content requirements and the dynamic occupation of spectrum by primary users. Previous DRL approaches for caching often treat multi-cache systems as homogeneous agents or model the entire system as a single global agent. This overlooks the fundamental aspect that users and cache servers (SBSs) are autonomous agents with potentially conflicting objectives: users seek minimum delivery delay, while SBSs aim to maximize reputation or revenue by storing popular content. Specifically, the paper addresses how cache nodes can maximize storage efficiency, how users can choose the best cache node for minimal delay, and how to design an effective learning algorithm for heterogeneous agents operating in a mixed cooperative-competitive environment.

4. Assumptions

- There is one Primary Base Station (PBS) that contains all contents.
- There are N Secondary Base Stations (SBSs) acting as edge cache servers, each with a caching size C_n .
- There are K users in the system.
- Each user and SBS is equipped with a single omni-directional antenna.
- An 'SBS-first' policy is adopted, meaning users first attempt to fetch content from an appropriate SBS, and if not available, they access the PBS.
- Cooperative caching of popular contents in SBSs occurs during the off-peak time of the primary network.
- The transmission rate of SBSs ($r_{n,k}$) is faster than that of PBS ($r_{o,k}$).
- The slot duration (T_o) is long enough for all content requests of users to be served within a single time slot.
- The overall contents consist of F files, and their file sizes (s_f) are independently and identically distributed (i.i.d) according to a Pareto distribution with an average value of 5.
- The contents request vector ($q_k(t)$) follows a Zipf distribution.
- Contents are modeled as static, but the number of requests (q_k,f) is time-variant with respect to users.
- Each file can be divided into fragments using a rateless code (e.g., Raptor codes).
- For user k , it can only retrieve one fragment of file f from a single SBS n with size $A_{n,f}$, and the remaining fraction $(1-A_{n,f})$ from PBS, rather than employing multiple SBSs to cooperatively relay fragments.
- SBSs are geographically closer to users than PBS, leading to better transmission rates and less delivery delay.

5. Limitations

- The content popularity is modeled as static, although content requests are time-variant, which might not fully capture real-world dynamics.
- The model simplifies content fetching by assuming a user can only retrieve one fragment of a file from a single SBS, with the rest from the PBS, rather than allowing cooperative relaying of fragments from multiple SBSs.
- While decentralized execution reduces communication overhead, the centralized training aspect of MADDPG still requires global state and action information, which can be a limitation in extremely large-scale or highly dynamic environments.

6. Results and Conclusion

The numerical simulations demonstrate the effectiveness of the proposed heterogeneous MADDPG algorithm. Both user and SBS rewards successfully converge during training, confirming the algorithm's stability. The results show that SBSs' rewards are influenced by their geographic location, with farther SBSs adopting different caching strategies to attract users. The replacement cost factor (β) significantly impacts SBS storage strategy, with higher costs leading to a stronger preference for caching more popular content, especially under storage constraints. Crucially, the heterogeneous MADDPG approach consistently outperforms homogeneous MADDPG variants (where only users or SBSs learn) in terms of overall system reward, indicating its superior ability to balance cooperative-competitive relationships. Furthermore, it achieves significantly higher global rewards compared to other actor-critic based off-policy DRL algorithms like DDPG, PPO, and TD3, with a reported improvement of nearly 40% in total system reward. The final conclusion is that the proposed MADDPG effectively optimizes both the content storage strategy of SBSs and the content delivery delay of users simultaneously, leading to enhanced system performance in CRNs.

7. Real World Impact

This research has practical real-world applications in improving the performance of wireless networks, particularly in scenarios involving edge caching and dynamic spectrum access. It can lead to enhanced spectrum efficiency and alleviation of spectrum scarcity in Cognitive Radio Networks. By optimizing content delivery, it can significantly reduce the delivery delay for users, improving their quality of experience. Furthermore, by enhancing the cache efficiency of Secondary Base Stations (SBSs), it can lead to more effective resource utilization and potentially increased revenue for network operators. This approach is relevant for future edge AI and smart city applications where distributed, autonomous agents need to cooperate and compete for optimal system performance.

8. Graphs and Figures Analysis

Figure 1: System model

This figure illustrates the overall system architecture and the operational flow of the proposed heterogeneous Multi-Agent Deep Deterministic Policy Gradient (MADDPG) approach. The top part shows the network topology, with a Primary Base Station (PBS) serving multiple Secondary Base Stations (SBSs), which in turn serve multiple users. Users prioritize fetching content from SBSs (secondary access) and resort to PBS (primary access) if content is unavailable at SBSs. The bottom part details the multi-agent learning model, depicting K user agents and N SBS agents. Each agent has an 'Actor' network (π) responsible for generating actions and a 'Critic' network (Q) for evaluating actions. During the 'Training' phase (red data flow), critics use global observations (S_All) and global actions (A_All) to learn and guide the actors. During the 'Execution' phase (green data flow), well-trained actors make optimal actions independently based on their partial observations, reducing communication overhead.

TABLE I: Neural Network Configuration of MADDPG

This table provides the detailed configuration of the neural networks used for the Actor and Critic components within the MADDPG framework. For both user and SBS agents, the Actor network consists of an input layer (F neurons, ReLU activation), two hidden layers (300 neurons each, ReLU), and an output layer (F neurons, Sigmoid). The Critic network for both agent types has an input layer ($2(N+K)F$ neurons, ReLU), two hidden layers (1024 neurons each, ReLU), and a single output neuron (scalar value, no specific activation function mentioned as it's a policy gradient value scaler). The critic's input is the combined action-state pair (S_All, A_All).

Fig. 2: Locations and rewards of heterogeneous agents

This figure visualizes the simulation setup and demonstrates the convergence of rewards for the heterogeneous agents. Figure 2(a) 'Position Distribution' shows the spatial arrangement of the PBS (at origin), 5 SBSs (SBS1-SBS5), and 3 users (User1-User3) in a 2D coordinate system, illustrating the network topology. Figure 2(b) 'Users' reward' plots the individual rewards for each of the three users over training episodes, showing that all user rewards converge to stable negative values (around -2 to -3), indicating successful learning to minimize delivery delay. User 3 consistently achieves a slightly better (less negative) reward. Figure 2(c) 'SBSs' reward' plots the individual rewards for each of the five SBSs over training episodes, showing that all SBS rewards converge to stable positive values (around 1 to 3), demonstrating successful learning to optimize cache efficiency and revenue. SBS 1, 3, and 5 achieve higher rewards than SBS 2 and 4, which are geographically farther from users.

Fig. 3: Contents popularity vs. cached fraction

This figure illustrates how content popularity and the replacement cost factor (β) influence the caching strategy of SBSs. Figure 3(a) 'Impact by location of SBS' shows the average cached fraction (λ) versus normalized content popularity for each of the 5 SBSs. SBS 1, 3, and 5, being closer to users, exhibit similar caching curves, prioritizing higher fractions of more popular content. In contrast, SBS 2 and 4, located farther from users, show different curves, caching a higher fraction of even less popular content to compensate for their less favorable location and attract users. The shaded area indicates the storage capacity limit. Figure 3(b) 'Impact by factor β ' shows the average cached fraction (λ) versus normalized content popularity for varying replacement cost factors ($\beta = 0.5, 0.8, 1.0, 1.2$). As β increases, the curves transition from up-concave to down-convex, indicating that SBSs increasingly prioritize caching more popular content when replacement costs are higher or cache space is limited, demonstrating an adaptive storage strategy.

Fig. 4: Comparison with homogeneous MADDPG

This figure compares the overall rewards of users and SBSs across different learning scenarios, highlighting the benefits of the heterogeneous MADDPG. Figure 4(a) 'Users' overall reward' plots the sum of user rewards (ΣR_k) over episodes for three cases: heterogeneous MADDPG (both users and SBSs learn), homogeneous MADDPG (only SBSs learn, users use static fetching), and homogeneous MADDPG (only users learn, SBSs use LFU replacement). The heterogeneous MADDPG achieves the highest (least negative) user reward, indicating superior delivery delay minimization. Figure 4(b) 'SBSs' overall reward' similarly plots the sum of SBS rewards (ΣR_n) for the same scenarios. Here, the heterogeneous MADDPG achieves the highest positive SBS reward, signifying better cache efficiency and revenue. Both plots demonstrate that allowing both users and SBSs to learn cooperatively and competitively (heterogeneous MADDPG) yields significantly better performance for both user satisfaction and SBS efficiency compared to scenarios where only one type of agent learns or uses simplified strategies.

Fig. 5: Performance comparison with other DRL

This figure presents a performance comparison of the proposed heterogeneous MADDPG with other state-of-the-art actor-critic based off-policy Deep Reinforcement Learning (DRL) algorithms, including DDPG, PPO, and TD3. The y-axis represents the total global reward (sum of user and SBS rewards), and the x-axis represents training episodes. The plot clearly shows that the heterogeneous MADDPG consistently achieves the highest global reward, converging to a significantly higher value than all other compared DRL algorithms. This demonstrates the superior ability of the proposed approach to optimize the combined objectives of user delivery delay and SBS cache efficiency, achieving up to a 40% improvement in total system reward compared to homogeneous learning approaches.